

Beyond Seen Mistakes: A Matched Test of Natural Error-Composition Generalization in Exercise Assessment

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Abstract—Automated exercise assessment predicts several biomechanical criteria at once, yet grouped evaluation does not isolate whether a familiar diagnosis was present in training. We introduce a recording-pair-disjoint matched test on public ALEX-GYM-1 poses. For each of 12 natural diagnoses, target-present and target-absent models use identical test recordings, validation data, initialization, class weights, and training-set size. Across four temporal backbones—TCN, BiGRU, Transformer, and state-space—an optimized exchange minimizes residual criterion-marginal differences. Removing target-diagnosis examples then lowers exact diagnosis match from 22.65% to 1.52% (target bootstrap gap 21.13 points, 95% CI [9.86, 34.73]; two-sided $p = .00049$); bit accuracy falls 12.80 points. Every target falls, and the largest gaps occur where residual imbalance is smallest. A separate six-model full-target stress test, adding ST-GCN and an anatomical-token model, shows that topology and locality do not eliminate the broader failure. A label-space audit finds that one-bit opposing neighbors predict vulnerable criteria across models, but a far simpler statistic—the raw count of training examples opposing a criterion’s target state—is a stronger predictor ($\rho = -.742$) and absorbs the local count’s association entirely ($p = .200$ once controlled), separating an 82.70% bottom-quartile criterion from a 40.79% top-quartile one. The contribution is therefore a controlled composition test, an evidence-bounded diagnosis of its failure, and confirmation—not introduction—of a known imbalance signal that flags which criteria of a planned diagnosis are likely to fail before training.

Index Terms—exercise assessment, multi-label classification, compositional generalization, skeleton sequences, label-space diagnostics

I. INTRODUCTION

Fine-grained exercise assessment differs from ordinary action recognition. A system must report several simultaneous errors—for example foot orientation, hip position, back posture, and movement depth—rather than one action class. Consequently, a label is a binary diagnosis vector $\mathbf{y} \in \{0,1\}^C$, and the practically relevant question is whether individually familiar criterion states remain recognizable when they occur in a new combination.

The distinction is easy to miss. A recording-disjoint random split protects against source overlap but does not deliberately hold out a chosen diagnosis vector, so a model can obtain reasonable marginal accuracy while using pose patterns correlated with entire diagnoses. This matters for feedback: one systematically wrong criterion invalidates an otherwise plausible diagnosis.

Composite errors have been studied directly by CPR-Coach, which trains on isolated error classes and tests synthetically enumerated composites [1]. Held-out label combinations are also established in multi-label learning [2]. Our setting is narrower and naturalistic: training remains multi-error, every evaluated criterion state has training support, and test diagnoses are naturally observed in a public two-view exercise dataset, so we do not claim the first composite-error task or combination split.

We organize the study around two retrospective questions. **RQ1** asks whether removing a target diagnosis from training reduces performance on the same held-out recording groups under matched training size and loss weights. **RQ2** asks whether local one-bit label structure identifies vulnerable criteria beyond ordinary marginal criterion support. Tests are two-sided; exact match was fixed as the sole primary endpoint before the corrected matched rerun.

We contribute (i) a data-audited observed-composition protocol, (ii) a controlled target-present/absent comparison, (iii) a six-architecture stress test, and (iv) evidence that a plausible local label diagnostic is subsumed by ordinary per-criterion class imbalance, confirming a known imbalance-generalization relationship under composition shift and usable as a training-free check before any model is touched. Figure 1 shows the naturally observed diagnosis spaces and targets retained by the support rule.

II. RELATED WORK

A. Exercise assessment and composite errors

ALEX-GYM-1 provides two-view 3D poses and multi-criterion exercise annotations [3]. Other rehabilitation datasets emphasize clinical scores, correctness, or multi-task assessment rather than natural held-out multi-error label sets [4]–[6]. FineDiving and FineParser study procedure-aware action quality at fine temporal granularity [7], [8], but not this diagnosis-set shift. CPR-Coach is the closest direct composition formulation [1]; its isolated-error training regime differs from our naturally multi-error training pool.

Learnable Motion-Focused Tokenization drops low-motion RGB patches to reduce background-driven video domain shift [9]; our pose input has no scene background, so this is an adjacent evidence-selection principle, not a matched baseline here.

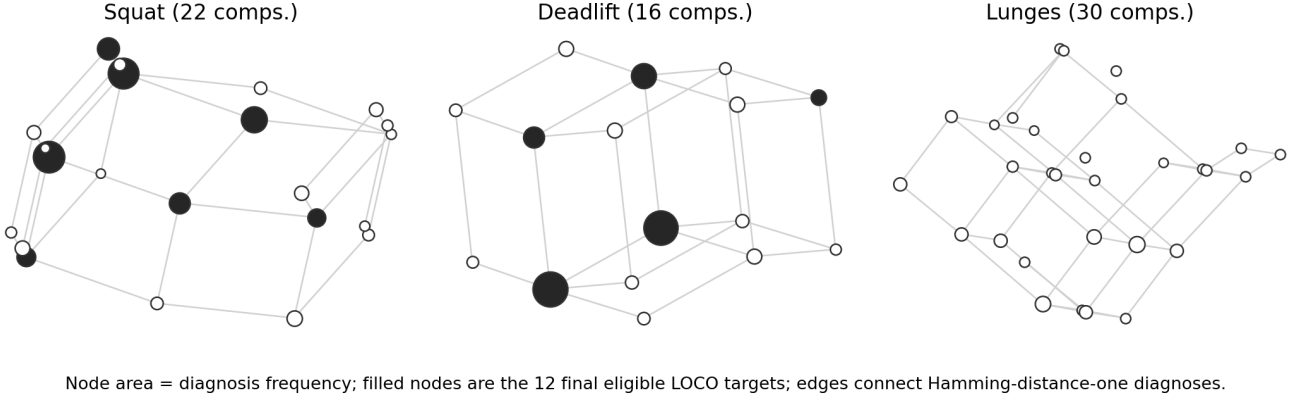


Fig. 1. Naturally observed diagnosis spaces. Node area represents repetition frequency, filled nodes are the 12 targets retained by the final support rule, and edges join diagnoses differing in exactly one criterion. Lunge has no filled node because its candidate targets cannot preserve the required train/validation state support after recording-group removal.

TABLE I
PROCESSED ALEX-GYM-1 DATA. “GROUPS” ARE PAIRED RECORDING IDENTIFIERS, NOT VERIFIED PARTICIPANT IDENTITIES.

Exercise	Repetitions	Criteria	Diagnoses	Groups
Squat	294	6	22	59
Deadlift	269	5	16	55
Lunge	106	7	30	53

B. Multi-label shortcuts and unseen combinations

Multi-label models often exploit label dependence. Compositional multi-label classification explicitly holds out label combinations [2], while counterfactual and correlation-shortcut work studies harmful co-occurrence dependence [10], [11]. We use this literature to motivate a diagnostic, not a causal claim: label counts characterize exposure in the benchmark and never enter the exercise classifier. Per-criterion opposing-state support is ordinary class imbalance, linked to degraded minority-state generalization [12]–[14]; our contribution is showing it subsumes a more localized label-neighborhood account under composition shift.

III. DATA AND PROTOCOL

A. Dataset and preprocessing audit

The public ALEX-GYM-1 subset contains squat, Romanian deadlift, and lunge repetitions with frontal and lateral 33-joint 3D pose sequences. We convert each trainer criterion to an error bit: one denotes a violated criterion. Table I summarizes the processed data.

All-zero frames occur in both views: squat has 183/4,720 frontal (3.88%) and 702/4,720 lateral frames (14.87%); deadlift has 250/4,304 (5.81%) and 326/4,304 (7.57%); and lunge has 33/1,696 (1.95%) and 214/1,696 (12.62%), respectively. We linearly interpolate whole missing frames before pelvis centering, robust body-scale normalization, and resampling to 16 frames. One squat repetition has no valid frontal frame and is excluded. Raw-file SHA-256 hashes, the exclusion, and

per-sequence quality counts are released. Isolated zero joints are not imputed, since their semantics cannot be verified.

B. Observed-composition holdout

For exercise e , let s_i identify a paired frontal/lateral recording and y_i its diagnosis. For target diagnosis y^* , the test set is all repetitions with $y_i = y^*$. Every recording group containing that target is removed from the train/validation pool, including its non-target repetitions. The remaining groups are split deterministically. Candidate splits are rejected unless both states of every criterion occur at least eight times in training and at least once in validation. Train-validation, train-test, and validation-test group intersections are zero in every fold.

A target is eligible when it has at least ten repetitions, three source groups, and a feasible supported split. Seven squat and five deadlift diagnoses qualify. Two lunge diagnoses considered by the earlier protocol are infeasible: their minority pool contains exactly eight examples, so eight cannot remain in training while one also enters validation. They are excluded rather than evaluated under a different rule. The primary experiment is therefore 12 targets $\times$ three seeds per model.

Ordinary validation is a secondary comparator, not a pure seen-composition control: 409 of 1,760 validation rows (23.2%) have a diagnosis absent from that fold’s training subset.

C. Matched composition-presence test

RQ1 instead holds the evaluation data fixed. For each target and seed, one target recording group is test-only; the three seeds select distinct groups. Validation uses groups containing no target repetition. The *present* training set contains target repetitions from the remaining target groups. The *absent* set exchanges those rows for an equal number of non-target rows from the same training-group pool. Because every eligible replacement differs from the target in at least one bit, exact criterion-marginal equality is impossible. Before training, a deterministic binary optimization therefore minimizes the largest per-criterion positive-count difference and then their sum. Relative to random exchange, their fold-mean maximum

falls from 22.06 to 10.89 counts and their total from 55.08 to 33.42. Thus the conditions have identical size, validation and test rows, initialization, shared class weights, and zero group overlap, with reduced but nonzero marginal differences. This test uses four generic temporal backbones and averages seeds, then models, within each diagnosis.

D. Models and training

We use six compact controlled models: a temporal CNN, BiGRU, Transformer, selective state-space model, ST-GCN, and Factorized Anatomical Criterion Tokens (FACT). All baselines receive a 16×198 sequence (two views $\times$ 33 joints $\times$ xyz). The TCN uses a 96-D frame embedding, temporal kernels 5 and 3, and mean/max pooling. BiGRU uses a 72-D bidirectional GRU; Transformer uses two 96-D, four-head encoder layers; and SSM uses one 32-D bidirectional selective scan. Their deadlift models contain 112k, 99k, 190k, and 13k parameters, respectively. The SSM is a compact study implementation, not an official reproduction. ST-GCN applies two spatial graph/temporal convolution blocks to each MediaPipe view using the physical skeleton edges, then fuses pooled view features (35k deadlift parameters).

FACT reshapes each sequence to $2 \times 16 \times 33 \times 3$. For criterion c , the annotation wording specifies one view and an initial joint set. A learned residual adjusts a sigmoid mask initialized near .95 on mapped joints and .05 elsewhere. Separate frontal/lateral encoders apply a 56-D frame projection, temporal convolutions of width 5 and 3, and mean/max pooling. A criterion-specific $112 \rightarrow 32$ adapter produces one token per criterion, whose logit sums a local linear score and a scaled distance to two learned 32-D state prototypes. FACT never consumes other labels at inference and has 84,858 squat and 81,048 deadlift parameters. It is a locality-oriented comparator, not a claimed superior architecture.

Every model receives the same representation and split, with class-weighted binary cross-entropy, AdamW, fixed 0.5 thresholds, and early stopping on validation macro-F1. Hyperparameters are never selected using the held-out target. We report target-equal means: seeds are averaged within target, then all 12 targets receive equal weight. All primary trainers call the same deterministic constrained splitter. The released manifest records raw-file hashes, preprocessing version, exact indices, target support, exclusions, and zero group intersections for every target and seed.

IV. LABEL-SPACE DIAGNOSTIC

For criterion c , let $\mathbf{y}_{-c}^*$ denote the other criterion bits. Label-Opposition Pressure is

$$LOP_c(\mathbf{y}^*) = \sum_{i \in \mathcal{T}} \mathbb{I}[\mathbf{y}_{i,-c} = \mathbf{y}_{-c}^*] \mathbb{I}[y_{i,c} \neq y_c^*], \quad (1)$$

where $\mathcal{T}$ is the fold training set. Large pressure means many training examples reproduce the target context while supporting the wrong current bit. We compare it with global opposing-state support, $G_c = \sum_{i \in \mathcal{T}} \mathbb{I}[y_{i,c} \neq y_c^*]$, a single training-label count requiring no context matching. Target frequency, cardinality,

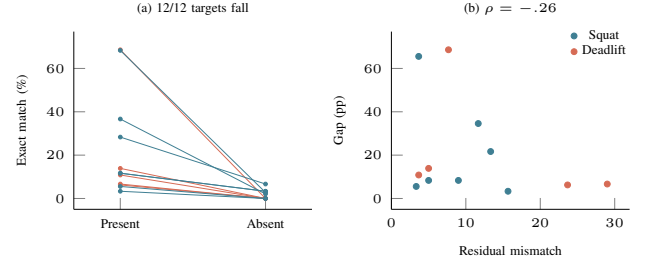


Fig. 2. (a) Exact diagnosis match for each of the 12 held-out diagnoses, under matched training size, class weights, initialization, validation and test rows. (b) The same gaps against the residual worst-criterion mismatch the exchange optimization could not remove. Color marks exercise.

and total one-bit neighborhood density are constants across criteria inside a diagnosis and cancel under within-diagnosis ranking. Both LOP and G_c are exposure associations, not causal influence or a guarantee of failure.

V. STATISTICAL ANALYSIS

The held-out diagnosis is the statistical unit. For RQ1, we average seeds and the four matched architectures within target, then test 12 present-absent differences. Exact match is primary; bit accuracy and F1 are secondary. For RQ2, we average all five original temporal models over seeds, compute Spearman correlation within diagnosis, and take the raw diagnosis-equal mean across 67 target-criterion units. A 50,000-shuffle test permutes a predictor only among criteria of the same diagnosis. A partial rank test residualizes LOP and accuracy against G_c and target-bit state before applying the same blocked permutation. We report 100,000-sample target bootstrap intervals and two-sided probabilities. Exact match was fixed before both corrected reruns; the exchange optimization was specified before examining its model outcomes.

VI. RESULTS

A. RQ1: matched removal exposes diagnosis dependence

Under optimized exchange and matched test recordings, target-present exact match is 22.65%, versus 1.52% when the target is absent from training: a 21.13-point gap (95% CI [9.86, 34.73]; two-sided Wilcoxon $p = .00049$). The gap is positive for every architecture (26.76 TCN, 20.69 BiGRU, 20.14 Transformer, and 16.94 SSM points). Bit accuracy also falls 12.80 points (CI [7.38, 18.79]; $p = .00049$). The original random-exchange analysis gives a compatible 25.44-point exact-match gap (CI [17.31, 33.32]). RQ1 is therefore supported without comparing different evaluation distributions and is robust to substantially improved marginal balance.

Figure 2 resolves the 12 targets individually. All 12 fall, and residual imbalance does not account for the effect: the two largest gaps (68.61, 65.56 points) sit at two of the smallest residual mismatches (7.7, 3.7 counts), and the gap-mismatch rank association is negative ($\rho = -.26$), opposite the confound's prediction. At 12 targets this is suggestive, not decisive.

The secondary full-target stress test still measures a harder distributional setting. Table II shows that FACT has the largest

TABLE II
TARGET-EQUAL PERFORMANCE (%). VALIDATION IS ORDINARY
GROUP-HELD-OUT; LOCO DELIBERATELY WITHHOLDS ONE NATURAL
DIAGNOSIS.

Model	Validation			LOCO		
	Bit	μ F1	Exact	Bit	μ F1	Exact
TCN	65.08	56.35	12.61	63.04	44.86	2.39
BiGRU	65.32	57.00	14.71	60.61	42.11	2.63
Transformer	64.26	57.08	12.20	59.83	41.79	2.07
Selective SSM	66.89	56.91	14.23	62.50	41.82	3.41
ST-GCN	64.61	56.08	10.86	65.87	49.82	2.85
FACT	72.39	63.95	18.56	64.89	44.77	5.14

mean bit accuracy and exact match, but its advantage over TCN is unresolved: +1.85 bit points ($p = .204$) and +2.75 exact points ($p = .086$) in two-sided exploratory comparisons. We therefore make no architecture-superiority claim. Micro-F1 is pooled across criterion decisions; macro-F1 is omitted because every criterion has one constant state inside a target fold, making it meaningless here. “Bit” is the fraction of all criterion decisions that are correct; “Exact” requires every criterion of a repetition to be correct—so FACT’s 64.89% LOCO Bit but 5.14% Exact means most individual decisions are right while few complete diagnoses are.

B. RQ2: marginal support, not local opposition, is the practical predictor

Raw LOP is negatively associated with held-out criterion accuracy across all five temporal models (per-model ρ from $-.367$ to $-.648$, all blocked $p \leq .00584$); the model-mean ρ is $-.632$ ($p < .00004$). However, global opposing-state support is stronger ($\rho = -.742$, $p < .00004$). After controlling for that support and the target bit, LOP’s partial rank association is $-.182$ ($p = .200$). RQ2 is therefore not supported in its incremental form: the local count is descriptive, but these data favor the simpler marginal-support account.

The association is practically large: among the 67 target-criterion units, the bottom G_c quartile (least opposed) averages 82.70% held-out accuracy versus 40.79% for the top quartile (most opposed). G_c is an ordinary imbalance count over training labels, so this confirms—rather than introduces—a training-free audit signal for a proposed diagnosis composition, available without running an experiment.

VII. DISCUSSION

The matched comparison changes the evidential strength: target presence is varied while evaluation recordings and training budget are fixed. The effect appears in convolutional, recurrent, attention, and state-space models, and ST-GCN’s low full-target exact match shows explicit skeleton topology is not by itself a remedy.

The control analysis also changes the mechanism claim. LOP reliably describes which bit fails, but it is largely a proxy for ordinary criterion-state imbalance (Sec. II-B). The practical takeaway is therefore not a label-neighborhood mechanism but that plain imbalance, on its own, predicts held-out failure well enough to serve as a deployment-time check requiring no training run.

VIII. LIMITATIONS

The study has one multi-label composition dataset and only 12 eligible diagnosis units. The split key identifies paired recordings, not verified participants. The matched test uses three test groups per diagnosis and exchanges examples rather than preserving identical marginals; optimization roughly halves the mean worst-criterion difference, but residual imbalance prevents a causal interpretation. Manual criterion–anatomy mappings affect FACT, which is why it is only a comparator. Thresholds are fixed at 0.5, with no calibrated-probability claim. Missing-frame rates reach 14.87% in the squat lateral view; interpolation reduces but cannot eliminate pose-detector uncertainty. The G_c quartile split (17 units/quartile) reuses the RQ2 sample, so it is descriptive, not independent validation. Exact match is clinically intuitive but harsh; we also report bit accuracy and micro-F1.

IX. CONCLUSION

On identical held-out recordings, removing a natural diagnosis from matched training reduces exact match by 21.13 points across four architecture families, even after criterion-marginal differences are substantially reduced. ST-GCN and anatomical tokens do not eliminate the broader stress-test gap. Label-space analysis predicts vulnerable criteria, and ordinary training-label imbalance—not local one-bit opposition—does the predicting. The result is a reproducible composition test, an evidence-bounded diagnosis of its failure, and confirmation that this known imbalance signal, not label-neighborhood structure, is what predicts failure here—a training-free check worth running before a model is trained.

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